BEARING CAPACITY

rectangle isolated footing

PARAMETERS

$h_{cover} := 0.7\ m$

$B := 1.1\ m$

$L := 0.9\ m$

$\alpha := 77\ deg$

$h_f := 0.8\ m$

$b_{cover} := 0.4\ m$

$l := 0.4\ m$

$G_K := 102\ kN$

$Q_K := 75\ kN$

$h_w := 4.46\ m$

Soil layer 1

$\gamma_{1unsat} := 18.5\ \frac{kN}{m^3}$

$\gamma_{1sat} := 19.5\ \frac{kN}{m^3}$

$\varphi_1 := 32\ deg$

$c_1 := 0\ kPa$

$cu_1 := 0\ kPa$

$E_1 := 8\ MPa$

$h_1 := 1.5\ m$

Rock

$E_R := 300\ MPa$

$\gamma_d := 25\ \frac{kN}{m^3}$

$\gamma_w := 10\ \frac{kN}{m^3}$

Soil layer 2

$\gamma_{2unsat} := 19\ \frac{kN}{m^3}$

$\gamma_{2sat} := 20.5\ \frac{kN}{m^3}$

$\varphi_2 := 19\ deg$

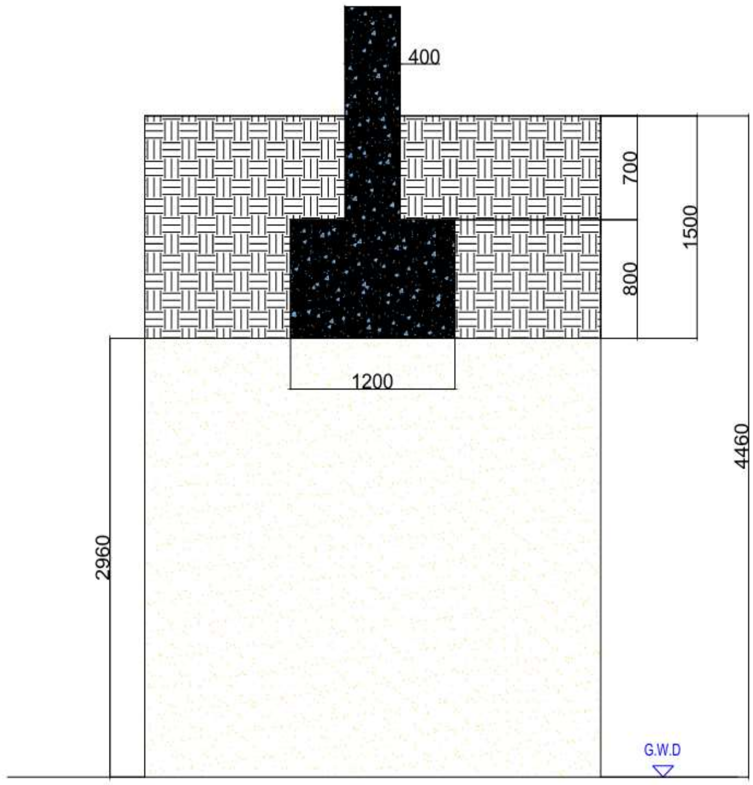
$c_2 := \blacksquare\ kPa$

$cu_2 := \blacksquare\ kPa$

$E_2 := 21\ MPa$

$h_2 := 7\ m$

the soil layer number 2 and the rock aren't used since the foundation touches the end of layer 1



$G_{vk} := G_K \cdot \sin(\alpha) = 99.3857\ kN$

$G_{hk} := G_K \cdot \cos(\alpha) = 22.945\ kN$

$Q_{vk} := Q_K \cdot \sin(\alpha) = 73.0778\ kN$

$Q_{hk} := Q_K \cdot \cos(\alpha) = 16.8713\ kN$

$e_{bk} := \frac{\left(G_{hk} + Q_{hk}\right) \cdot h_f}{\left(G_{vk} + Q_{vk}\right)} = 0.1847\ m$

$t_{wd} := \left(h_w - \left(h_f + h_{cover}\right)\right) = 2.96\ m$

$\boxed{B' := B - 2 \cdot e_{bk} = 0.7306\ m}$

$L' := L = 0.9\ m$

$\boxed{\gamma' := \gamma_{1unsat}}$

You only use γ' when:
1-The foundation is below the groundwater table, or
2-The failure surface dips into the saturated zone (for deeper/wider foundations).

Self of the footing

$G_f := B \cdot L \cdot h_f \cdot \gamma_d = 19.8\ \text{kN}$

Self f the cover soil

$G_{cover} := \left(B \cdot L - b_{cover} \cdot l\right) \cdot \gamma_{1unsat} \cdot h_{cover} = 10.7485\ \text{kN}$

Vertical n the footing

$V_d := \left(G_{vk} + G_f + G_{cover}\right) \cdot 1.35 + Q_{vk} \cdot 1.5 = 285.0279\ \text{kN}$

Effective static load at the bottom of the foundatiojn

$q := \gamma_{1unsat} \cdot h_1 + \gamma_{2unsat} \cdot \left(h_f + h_{cover} - h_1\right) = 27.75\ \frac{\text{kN}}{\text{m}^2}$

Shape elements

$S_q := 1 + \frac{B'}{L'} \cdot \sin\left(\varphi_1\right) = 1.4302$

$S_\gamma := 1 - 0.3 \cdot \frac{B'}{L'} = 0.7565$

$S_c := \frac{S_q \cdot N_q - 1}{\left(N_q - 1\right)} = 1.4496$

Load earing elements

$N_q := \mathbf{e}^{\boldsymbol{\pi} \cdot \tan\left(\varphi_1\right)} \cdot \left(\tan\left(45\ \text{deg} + \frac{\varphi_1}{2}\right)\right)^2 = 23.1768$

$N_\gamma := 2 \cdot \left(N_q - 1\right) \cdot \tan\left(\varphi_1\right) = 27.7152$

$N_c := \left(N_q - 1\right) \cdot \cot\left(\varphi_1\right) = 35.4903$

verticality of loading

$f := \frac{G_{hk}}{\left(G_{vk} + B' \cdot L' \cdot c_1 \cdot \cot\left(\varphi_1\right)\right)} = 0.2309$

$m := \frac{2 + \frac{B'}{L'}}{1 + \frac{B'}{L'}} = 1.5519$

$i_\gamma := \exp\left(\left(1 + m\right) \cdot \ln\left(\left|1 - f\right|\right)\right) = 0.5118$

$i_q := \exp\left(m \cdot \ln\left(\left|1 - f\right|\right)\right) = 0.6654$

$i_c := \frac{\left(i_q \cdot N_q - 1\right)}{\left(N_q - 1\right)} = 0.6503$

soil horizontality

$b_\gamma := 1$

$b_q := 1$

$b_c := b_\gamma = 1$

$$R_k := (1)\cdot(B'\cdot L')\cdot\left(S_\gamma\cdot\gamma'\cdot B'\cdot N_\gamma\cdot i_\gamma\cdot b_\gamma\cdot 0.5 + S_q\cdot q\cdot N_q\cdot i_q\cdot b_q + S_c\cdot c_1\cdot N_c\cdot i_c\cdot b_c\right) = 450.1348\ \mathrm{kN}$$

$$R_d := \frac{R_k}{1.4} = 321.5248\ \mathrm{kN}$$

$$Utilization := \frac{V_d}{R_d} = 88.6488\ \%$$

SLIDING

Self of the footing

$$G_f := B\cdot L\cdot h_f\cdot\gamma_d = 19.8\ \mathrm{kN}$$

Self f the cover soil

$$G_{cover} := \left(B\cdot L - b_{cover}\cdot l\right)\cdot\gamma_{1unsat}\cdot h_{cover} = 10.7485\ \mathrm{kN}$$

Vertical n the footing

$$V_{d2} := \left(G_f + G_{cover} + G_{vk}\right)\cdot 1.35 + Q_{vk}\cdot 1.5 = 285.0279\ \mathrm{kN}$$

Active soil pressure

$$K_{a1} := \left(\tan\left(45\ \mathrm{deg} - \frac{\varphi_1}{2}\right)\right)^2 = 0.3073$$

vertical pressure

$$V_{1top} := h_{cover}\cdot\gamma_{1unsat}\cdot L = 11655\ \mathrm{m\ Pa}$$

$$V_{1bottom} := h_1\cdot\gamma_{1unsat}\cdot L = 24975\ \mathrm{m\ Pa}$$

Horizontal pressure

$$H_{soil1} := \frac{V_{1top} + V_{1bottom}}{2}\cdot K_{a1}\cdot\left(h_1 - h_{cover}\right) = 4.502\ \mathrm{kN}$$

$$H_{Tsoil} := H_{soil1} = 4.502\ \mathrm{kN}$$

$$H_d := \left(G_{hk} + H_{Tsoil}\right)\cdot 1.35 + Q_{hk}\cdot 1.5 = 62.3604\ \mathrm{kN}$$

At rest soil pressure

$$K_{01} := 1 - \sin\left(\varphi_1\right) = 0.4701$$

$$H_{SO1} := \frac{V_{1top} + V_{1bottom}}{2}\cdot K_{01}\cdot\left(h_1 - h_{cover}\right) = 6887.6229\ \mathrm{N}$$

$$H_{TSO} := H_{SO1} = 6887.6229\ \mathrm{N}$$

Charactaristic vertical load

$$V'_C := G_{vk} + G_f + G_{cover} + Q_{vk} = 203.012\ \text{kN}$$

Resistance against sliding

$$R_{dsliding} := \frac{V'_C \cdot \tan\left(\varphi_2\right) + H_{TSO}}{1.1} = 69.8093\ \text{kN}$$

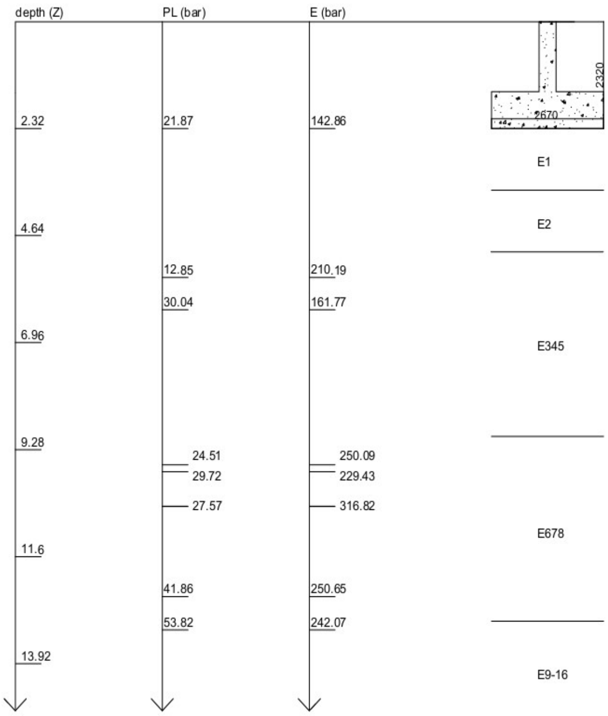
$$U := \frac{H_d}{R_{dsliding}} = 89.3296\ \%$$

here there was few possible solution like making the width = 2.8 m and the length= 6 m , angle 57°.the best possible solution is to tilt the superstructure to make the angle 77° and the B=1.2 m , L=0.9 m which will work fine with te low load that was assumed

Defing parameters

$L := 9.38\ \mathrm{m}$ $B := 2.67\ \mathrm{m}$ $D := 2.32\ \mathrm{m}$ $1.5 \cdot B = 4.005\ \mathrm{m}$

Normally consolidated sandy soil



Pressure	Depths	Elastic Modulus
$PL_1 := 21.87\ \mathrm{bar}$	$d_1 := 2.32\ \mathrm{m}$	$E_1 := 142.86\ \mathrm{bar}$
$PL_2 := 12.85\ \mathrm{bar}$	$d_2 := 5.55\ \mathrm{m}$	$E_2 := 210.19\ \mathrm{bar}$
$PL_3 := 30.04\ \mathrm{bar}$	$d_3 := 6.25\ \mathrm{m}$	$E_3 := 161.77\ \mathrm{bar}$
$PL_4 := 24.51\ \mathrm{bar}$	$d_4 := 9.61\ \mathrm{m}$	$E_4 := 250.09\ \mathrm{bar}$
$PL_5 := 29.72\ \mathrm{bar}$	$d_5 := 9.76\ \mathrm{m}$	$E_5 := 229.43\ \mathrm{bar}$
$PL_6 := 27.57\ \mathrm{bar}$	$d_6 := 10.51\ \mathrm{m}$	$E_6 := 316.82\ \mathrm{bar}$
$PL_7 := 41.86\ \mathrm{bar}$	$d_7 := 12.47\ \mathrm{m}$	$E_7 := 250.56\ \mathrm{bar}$
$PL_8 := 53.82\ \mathrm{bar}$	$d_8 := 13.19\ \mathrm{m}$	$E_8 := 242.07\ \mathrm{bar}$

Soil Class		Pl (MPa)	qc (MPa)
Clay and Silt	A	<0.7	<3.0
	B	1.2-2.0	3.0-6.0
	C	>2.5	>6.0
Sand and Gravel	A	<0.5	<5
	B	1.0-2.0	8.0-15.0
	C	>2.5	>20
Chalks	A	<0.7	<5
	B	1.0-2.5	>5
	C	>3.0	-
Marls and marly limestone	A	1.5-4.0	-
	B	>4.5	-
Rocks	A	2.5-4.0	-
	B	>4.5	-

Calculating the bearing capacity

$$PL1 := PL_1 = 2.187\ \mathrm{MPa}$$

from this we have sand and gravel C

$$PL2 := PL_1 = 2.187 \cdot 10^{6}\ \mathrm{Pa}$$

$$PLe := \left(PL1 \cdot PL2\right)^{\frac{1}{2}} = 2.187\ \mathrm{MPa}$$

$$D_e := \frac{1}{PLe} \cdot \left(D \cdot PL_1\right)$$

$$k_p := \left(1 + 0.8 \cdot \left(0.6 + 0.4 \cdot \frac{B}{L}\right) \cdot \frac{D_e}{B}\right)$$

$$q_a := \frac{1}{3} \cdot PLe \cdot k_p = 1.0907 \cdot 10^{6}\ \mathrm{Pa}$$

Soil Type	Expression of k_p
Clay and Silt A, Chalks A	$0{,}8.\left[1 + 0{,}25.\left(0{,}6 + 0{,}4\frac{B}{L}\right)\frac{D_e}{B}\right]$
Clay and Silt B	$0{,}8.\left[1 + 0{,}35.\left(0{,}6 + 0{,}4\frac{B}{L}\right)\frac{D_e}{B}\right]$
Clay C	$0{,}8.\left[1 + 0{,}5.\left(0{,}6 + 0{,}4\frac{B}{L}\right)\frac{D_e}{B}\right]$
Sand A	$\left[1 + 0{,}35.\left(0{,}6 + 0{,}4\frac{B}{L}\right)\frac{D_e}{B}\right]$
Sand and Gravel B	$\left[1 + 0{,}5.\left(0{,}6 + 0{,}4\frac{B}{L}\right)\frac{D_e}{B}\right]$
Sand and Gravel C	$\left[1 + 0{,}8.\left(0{,}6 + 0{,}4\frac{B}{L}\right)\frac{D_e}{B}\right]$
Chalks B and C	$1{,}3.\left[1 + 0{,}27.\left(0{,}6 + 0{,}4\frac{B}{L}\right)\frac{D_e}{B}\right]$
Marls, Marly limestones, Rocks	$\left[1 + 0{,}27.\left(0{,}6 + 0{,}4\frac{B}{L}\right)\frac{D_e}{B}\right]$

Calculating settlement

$\frac{L}{B} = 3.5131 \cdot \frac{1}{\text{m}}\ \text{m}$ $\qquad B_0 := 0.6\ \text{m}$

$\lambda_c := 1.40 \qquad \lambda_d := 2.14$

$\alpha := \frac{1}{3}$

$E_c := \frac{1}{\frac{1}{E_1}} = 142.86\ \text{bar}$

$E2 := E_1 = 142.86\ \text{bar}$

$E_{345} := \frac{2}{\frac{1}{E_2} + \frac{1}{E_3}} = 182.8285\ \text{bar}$

$E_{678} := \frac{4}{\frac{1}{E_4} + \frac{1}{E_5} + \frac{1}{E_6} + \frac{1}{E_7}} = 257.9877\ \text{bar}$

$E_{9.16} := E_8 = 242.07\ \text{bar}$

Rheological factor α (LCPC-SETRA, 1972)

Type	Peat	Clay		Silt		Sand		Gravel		Rock	
	α	E_M/p_l	α	E_M/p_l	α	E_M/p_l	α	E_M/p_l	α	Type	α
Overconsolidated or very tight		>16	1	>14	2/3	>12	1/2	>10	1/3	Very lowly fractured	2/3
Normally consolidated or normally tight	1	9–16	2/3	8–14	1/2	7–12	1/3	6–10	1/4	Normal	1/2
Under-consolidated weathered and remoulded or loose		7–9	1/2	5–8	1/2	5–7	1/3			Highly fractured	1/3
										Highly weathered	2/3

L/B	Circle	Square	2	3	5	20
λ_c	1.00	1.10	1.20	1.30	1.40	1.50
λ_d	1.00	1.12	1.53	1.78	2.14	2.65

$$\frac{4}{E_d} := \frac{1}{E_1} + \frac{1}{0.85 \cdot E2} + \frac{1}{E_{3.5}} + \frac{1}{2.5 \cdot E_{6.8}} + \frac{1}{2.5 \cdot E_{9.16}}$$

$$E_d := \left(\frac{4}{\frac{1}{E_1} + \frac{1}{0.85 \cdot E2} + \frac{1}{E_{345}} + \frac{1}{2.5 \cdot E_{678}} + \frac{1}{2.5 \cdot E_{9.16}}} \right) = 1.6731 \cdot 10^{7}\ \text{Pa}$$

$$\boxed{S_c := \frac{\alpha}{9 \cdot E_c} \cdot q_a \cdot \lambda_c \cdot B = 10.5704\ \text{mm}}$$

$$\boxed{S_d := \frac{2}{9 \cdot E_d} \cdot q_a \cdot B_0 \cdot \left(\lambda_d \cdot \frac{B}{B_0} \right)^{\alpha} = 18.4245\ \text{mm}}$$

$\boxed{S_f := S_c + S_d = 28.9948\ \text{mm}}$ $\qquad \boxed{S_f > 25\ \text{mm}}$

here we can see that this foundation isn't good since S surpassed the maximam allowable settlement which is 25mm

BEARING CAPACITY

wall footing

PARAMETERS

$h_{cover} := 0.7\ \mathrm{m}$

$B := 1.4\ \mathrm{m}$

$\alpha := 76\ \mathrm{deg}$

$h_f := 0.8\ \mathrm{m}$

$b_{cover} := 0.4\ \mathrm{m}$

$l := 0.4\ \mathrm{m}$

$G_K := 102\ \frac{\mathrm{kN}}{\mathrm{m}}$

$Q_K := 75\ \frac{\mathrm{kN}}{\mathrm{m}}$

$h_w := 4.46\ \mathrm{m}$

Soil layer 1

$\gamma_{1unsat} := 18.5\ \frac{\mathrm{kN}}{\mathrm{m}^3}$

$\gamma_{1sat} := 19.5\ \frac{\mathrm{kN}}{\mathrm{m}^3}$

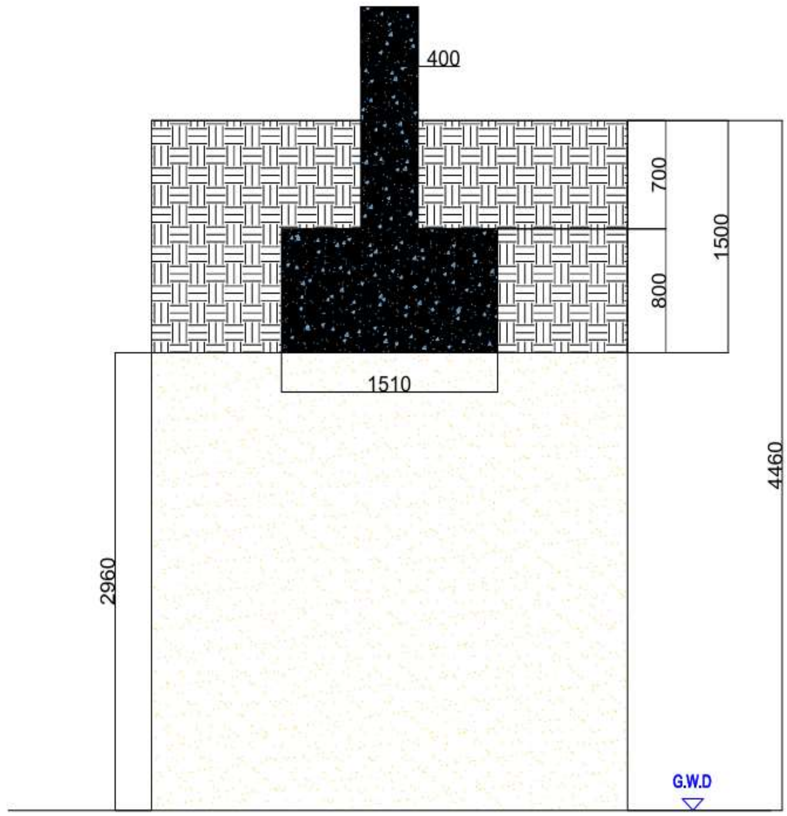
$\varphi_1 := 32\ \mathrm{deg}$

$c_1 := 0\ \mathrm{kPa}$

$cu_1 := 0\ \mathrm{kPa}$

$E_1 := 8\ \mathrm{MPa}$

$h_1 := 1.5\ \mathrm{m}$



Rock

$E_R := 300\ \mathrm{MPa}$

$\gamma_d := 25\ \frac{\mathrm{kN}}{\mathrm{m}^3}$

$\gamma_w := 10\ \frac{\mathrm{kN}}{\mathrm{m}^3}$

Soil layer 2

$\gamma_{2unsat} := 19\ \frac{\mathrm{kN}}{\mathrm{m}^3}$

$\gamma_{2sat} := 20.5\ \frac{\mathrm{kN}}{\mathrm{m}^3}$

$\varphi_2 := 19\ \mathrm{deg}$

$c_2 := \blacksquare\ \mathrm{kPa}$

$cu_2 := \blacksquare\ \mathrm{kPa}$

$E_2 := 21\ \mathrm{MPa}$

$h_2 := 7\ \mathrm{m}$

the soil layer number 2 and the rock aren't used since the foundation touches the end of layer 1

$G_{vk} := G_K \cdot \sin(\alpha) = 98.9702 \cdot \frac{1}{\mathrm{m}}\ \mathrm{kN}$

$G_{hk} := G_K \cdot \cos(\alpha) = 24.676 \cdot \frac{1}{\mathrm{m}}\ \mathrm{kN}$

$Q_{vk} := Q_K \cdot \sin(\alpha) = 72.7722 \cdot \frac{1}{\mathrm{m}}\ \mathrm{kN}$

$Q_{hk} := Q_K \cdot \cos(\alpha) = 18.1441 \cdot \frac{1}{\mathrm{m}}\ \mathrm{kN}$

$$e_{bk} := \frac{\left(G_{hk} + Q_{hk}\right) \cdot h_f}{\left(G_{vk} + Q_{vk}\right)} = 0.1995\ \mathrm{m}$$

$t_{wd} := \left(h_w - \left(h_f + h_{cover}\right)\right) = 2.96\ \mathrm{m}$

$\boxed{B' := B - 2 \cdot e_{bk} = 1.0011\ \mathrm{m}}$

$\boxed{\gamma' := \gamma_{1unsat}}$

Self of the footing

$$G_f := B \cdot h_f \cdot \gamma_d = 28 \cdot \frac{1}{\text{m}}\ \text{kN}$$

Self f the cover soil

$$G_{cover} := \left(B - b_{cover}\right) \cdot \gamma_{1unsat} \cdot h_{cover} = 12.95 \cdot \frac{1}{\text{m}}\ \text{kN}$$

Vertical n the footing

$$V_d := \left(G_{vk} + G_f + G_{cover}\right) \cdot 1.35 + Q_{vk} \cdot 1.5 = 298.0505 \cdot \frac{1}{\text{m}}\ \text{kN}$$

Effective static load at the bottom of the foundatiojn

$$q := \gamma_{1unsat} \cdot h_1 + \gamma_{2unsat} \cdot \left(h_f + h_{cover} - h_1\right) = 27.75\ \frac{\text{kN}}{\text{m}^2}$$

Shape elements

$$S_q := 1$$

$$S_\gamma := 1$$

$$S_c := 1$$

Load earing elements

$$N_q := \mathbf{e}^{\boldsymbol{\pi} \cdot \tan\left(\varphi_1\right)} \cdot \left(\tan\left(45\ \text{deg} + \frac{\varphi_1}{2}\right)\right)^2 = 23.1768$$

$$N_\gamma := 2 \cdot \left(N_q - 1\right) \cdot \tan\left(\varphi_1\right) = 27.7152$$

$$N_c := \left(N_q - 1\right) \cdot \cot\left(\varphi_1\right) = 35.4903$$

verticality of loading

$$f := \frac{G_{hk}}{\left(G_{vk} + B' \cdot L' \cdot c_1 \cdot \cot\left(\varphi_1\right)\right)} = 0.2493$$

$$m := 2$$

$$i_\gamma := \exp\left(\left(1 + m\right) \cdot \ln\left(\left|1 - f\right|\right)\right) = 0.423$$

$$i_q := \exp\left(m \cdot \ln\left(\left|1 - f\right|\right)\right) = 0.5635$$

$$i_c := \frac{\left(i_q \cdot N_q - 1\right)}{\left(N_q - 1\right)} = 0.5438$$

soil horizontality

$$b_\gamma := 1$$

$$b_q := 1$$

$b_c := b_\gamma = 1$

$R_k := (1) \cdot (B') \cdot \left(S_\gamma \cdot \gamma' \cdot B' \cdot N_\gamma \cdot i_\gamma \cdot b_\gamma \cdot 0.5 + S_q \cdot q \cdot N_q \cdot i_q \cdot b_q + S_c \cdot c_1 \cdot N_c \cdot i_c \cdot b_c \right) = 471.4917 \cdot \frac{1}{\text{m}} \text{ kN}$

$R_d := \frac{R_k}{1.4} = 336.7798 \cdot \frac{1}{\text{m}} \text{ kN}$

$$Utilization := \frac{V_d}{R_d} = 88.5001 \ \%$$

SLIDING

Self of the footing

$G_f := B \cdot h_f \cdot \gamma_d = 28 \cdot \frac{1}{\text{m}} \text{ kN}$

Self f the cover soil

$G_{cover} := \left(B - b_{cover} \right) \cdot \gamma_{1unsat} \cdot h_{cover} = 12.95 \cdot \frac{1}{\text{m}} \text{ kN}$

Vertical n the footing

$V_{d2} := \left(G_f + G_{cover} + G_{vk} \right) \cdot 1.35 + Q_{vk} \cdot 1.5 = 298.0505 \cdot \frac{1}{\text{m}} \text{ kN}$

Active soil pressure

$K_{a1} := \left(\tan\left(45 \text{ deg} - \frac{\varphi_1}{2} \right) \right)^2 = 0.3073$

vertical pressure

$V_{1top} := h_{cover} \cdot \gamma_{1unsat} = 12950 \text{ Pa}$

$V_{1bottom} := h_1 \cdot \gamma_{1unsat} = 27750 \text{ Pa}$

Horizontal pressure

$H_{soil1} := \frac{V_{1top} + V_{1bottom}}{2} \cdot K_{a1} \cdot \left(h_1 - h_{cover} \right) = 5.0022 \cdot \frac{1}{\text{m}} \text{ kN}$

$H_{Tsoil} := H_{soil1} = 5.0022 \cdot \frac{1}{\text{m}} \text{ kN}$

$H_d := \left(G_{hk} + H_{Tsoil} \right) \cdot 1.35 + Q_{hk} \cdot 1.5 = 67.2818 \cdot \frac{1}{\text{m}} \text{ kN}$

At rest soil pressure

$K_{01} := 1 - \sin\left(\varphi_1 \right) = 0.4701$

$H_{S01} := \frac{V_{1top} + V_{1bottom}}{2} \cdot K_{01} \cdot \left(h_1 - h_{cover} \right) = 7652.9144 \text{ m Pa}$

$$H_{TSO} := H_{SO1} = 7.6529 \ \frac{kN}{m}$$

Charactaristic vertical load

$$V'_C := G_{vk} + G_f + G_{cover} + Q_{vk} = 212.6923 \cdot \frac{1}{m} \ kN$$

Resistance against sliding

$$R_{dsliding} := \frac{V'_C \cdot \tan\left(\varphi_2\right) + H_{TSO}}{1.1} = 73.5352 \cdot \frac{1}{m} \ kN$$

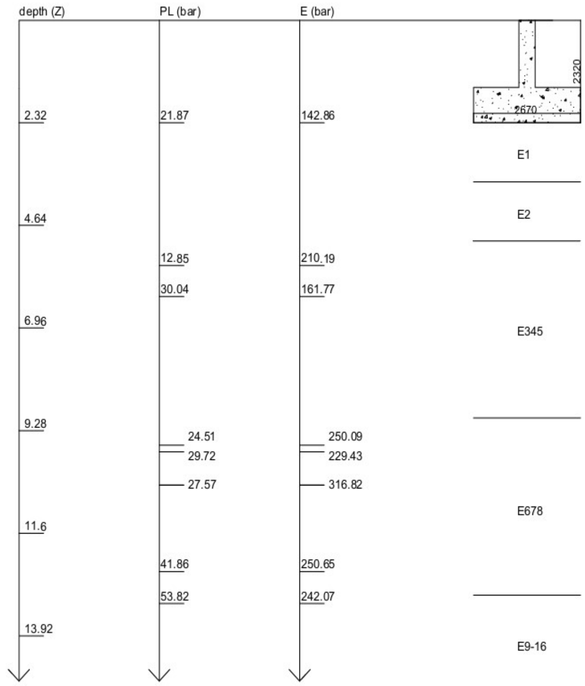
$$U := \frac{H_d}{R_{dsliding}} = 91.496 \ \%$$

here there was few possible solution like making the width = 2.8 m and the length= 6 m , angle 57°.the best possible solution is to tilt the superstructure to make the angle 77° and the B=1.2 m , L=0.9 m which will work fine with te low load that was assumed

Defing parameters

$L := 9.38\ \text{m}$ $\quad B := 2.67\ \text{m}$ $\quad D := 2.32\ \text{m}$ $\quad 1.5 \cdot B = 4.005\ \text{m}$

Normally consolidated sandy soil



Pressure	Depths	Elastic Modulus
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$PL_2 := 12.85\ \text{bar}$	$d_2 := 5.55\ \text{m}$	$E_2 := 210.19\ \text{bar}$
$PL_3 := 30.04\ \text{bar}$	$d_3 := 6.25\ \text{m}$	$E_3 := 161.77\ \text{bar}$
$PL_4 := 24.51\ \text{bar}$	$d_4 := 9.61\ \text{m}$	$E_4 := 250.09\ \text{bar}$
$PL_5 := 29.72\ \text{bar}$	$d_5 := 9.76\ \text{m}$	$E_5 := 229.43\ \text{bar}$
$PL_6 := 27.57\ \text{bar}$	$d_6 := 10.51\ \text{m}$	$E_6 := 316.82\ \text{bar}$
$PL_7 := 41.86\ \text{bar}$	$d_7 := 12.47\ \text{m}$	$E_7 := 250.56\ \text{bar}$
$PL_8 := 53.82\ \text{bar}$	$d_8 := 13.19\ \text{m}$	$E_8 := 242.07\ \text{bar}$

Soil Class		Pl (MPa)	qc (MPa)
Clay and Silt	A	<0.7	<3.0
	B	1.2-2.0	3.0-6.0
	C	>2.5	>6.0
Sand and Gravel	A	<0.5	<5
	B	1.0-2.0	8.0-15.0
	C	>2.5	>20
Chalks	A	<0.7	<5
	B	1.0-2.5	>5
	C	>3.0	-
Marls and marly limestone	A	1.5-4.0	-
	B	>4.5	-
Rocks	A	2.5-4.0	-
	B	>4.5	-

Calculating the bearing capacity

$$PL1 := PL_1 = 2.187\ \text{MPa}$$

from this we have sand and gravel C

$$PL2 := PL_1 = 2.187 \cdot 10^{6}\ \text{Pa}$$

$$PLe := (PL1 \cdot PL2)^{\frac{1}{2}} = 2.187\ \text{MPa}$$

$$D_e := \frac{1}{PLe} \cdot (D \cdot PL_1)$$

$$k_p := \left(1 + 0.8 \cdot \left(0.6 + 0.4 \cdot \frac{B}{L}\right) \cdot \frac{D_e}{B}\right)$$

$$q_a := \frac{1}{3} \cdot PLe \cdot k_p = 1.0907 \cdot 10^{6}\ \text{Pa}$$

Soil Type	Expression of k_p
Clay and Silt A, Chalks A	$0{,}8.\left[1+0{,}25.\left(0{,}6+0{,}4\frac{B}{L}\right)\frac{D_e}{B}\right]$
Clay and Silt B	$0{,}8.\left[1+0{,}35.\left(0{,}6+0{,}4\frac{B}{L}\right)\frac{D_e}{B}\right]$
Clay C	$0{,}8.\left[1+0{,}5.\left(0{,}6+0{,}4\frac{B}{L}\right)\frac{D_e}{B}\right]$
Sand A	$\left[1+0{,}35.\left(0{,}6+0{,}4\frac{B}{L}\right)\frac{D_e}{B}\right]$
Sand and Gravel B	$\left[1+0{,}5.\left(0{,}6+0{,}4\frac{B}{L}\right)\frac{D_e}{B}\right]$
Sand and Gravel C	$\left[1+0{,}8.\left(0{,}6+0{,}4\frac{B}{L}\right)\frac{D_e}{B}\right]$
Chalks B and C	$1{,}3.\left[1+0{,}27.\left(0{,}6+0{,}4\frac{B}{L}\right)\frac{D_e}{B}\right]$
Marls, Marly limestones, Rocks	$\left[1+0{,}27.\left(0{,}6+0{,}4\frac{B}{L}\right)\frac{D_e}{B}\right]$

Calculating settlement

$$\lambda_c := 2 \qquad B_0 := 0.6\ \text{m}$$

$$\lambda_d := 1.5$$

they are constant for wall

Rheological factor α (LCPC-SETRA, 1972)

	Peat	Clay		Silt		Sand		Gravel			Rock
Type	α	E_M/p_l	α	E_M/p_l	α	E_M/p_l	α	E_M/p_l	α	Type	α
Overconsolidated or very tight		>16	1	>14	2/3	>12	1/2	>10	1/3	Very lowly fractured	2/3
Normally consolidated or normally tight	1	9–16	2/3	8–14	1/2	7–12	1/3	6–10	1/4	Normal	1/2
Under-consolidated weathered and remoulded or loose		7–9	1/2	5–8	1/2	5–7	1/3			Highly fractured	1/3
										Highly weathered	2/3

L/B	Circle	Square	2	3	5	20
λ_c	1.00	1.10	1.20	1.30	1.40	1.50
λ_d	1.00	1.12	1.53	1.78	2.14	2.65

$$\alpha := \frac{1}{2}$$

$$E_c := \frac{1}{\frac{1}{E_1}} = 142.86\ \text{bar}$$

$$E2 := E_1 = 142.86\ \text{bar}$$

$$E_{345} := \frac{2}{\frac{1}{E_2} + \frac{1}{E_3}} = 182.8285\ \text{bar}$$

$$E_{678} := \frac{4}{\frac{1}{E_4} + \frac{1}{E_5} + \frac{1}{E_6} + \frac{1}{E_7}} = 257.9877\ \text{bar}$$

$$E_{9.16} := E_8 = 242.07\ \text{bar}$$

$$\frac{4}{E_d} := \frac{1}{E_1} + \frac{1}{0.85 \cdot E2} + \frac{1}{E_{3.5}} + \frac{1}{2.5 \cdot E_{6.8}} + \frac{1}{2.5 \cdot E_{9.16}}$$

$$E_d := \left(\frac{4}{\frac{1}{E_1} + \frac{1}{0.85 \cdot E2} + \frac{1}{E_{345}} + \frac{1}{2.5 \cdot E_{678}} + \frac{1}{2.5 \cdot E_{9.16}}} \right) = 1.6731 \cdot 10^{7}\ \text{Pa}$$

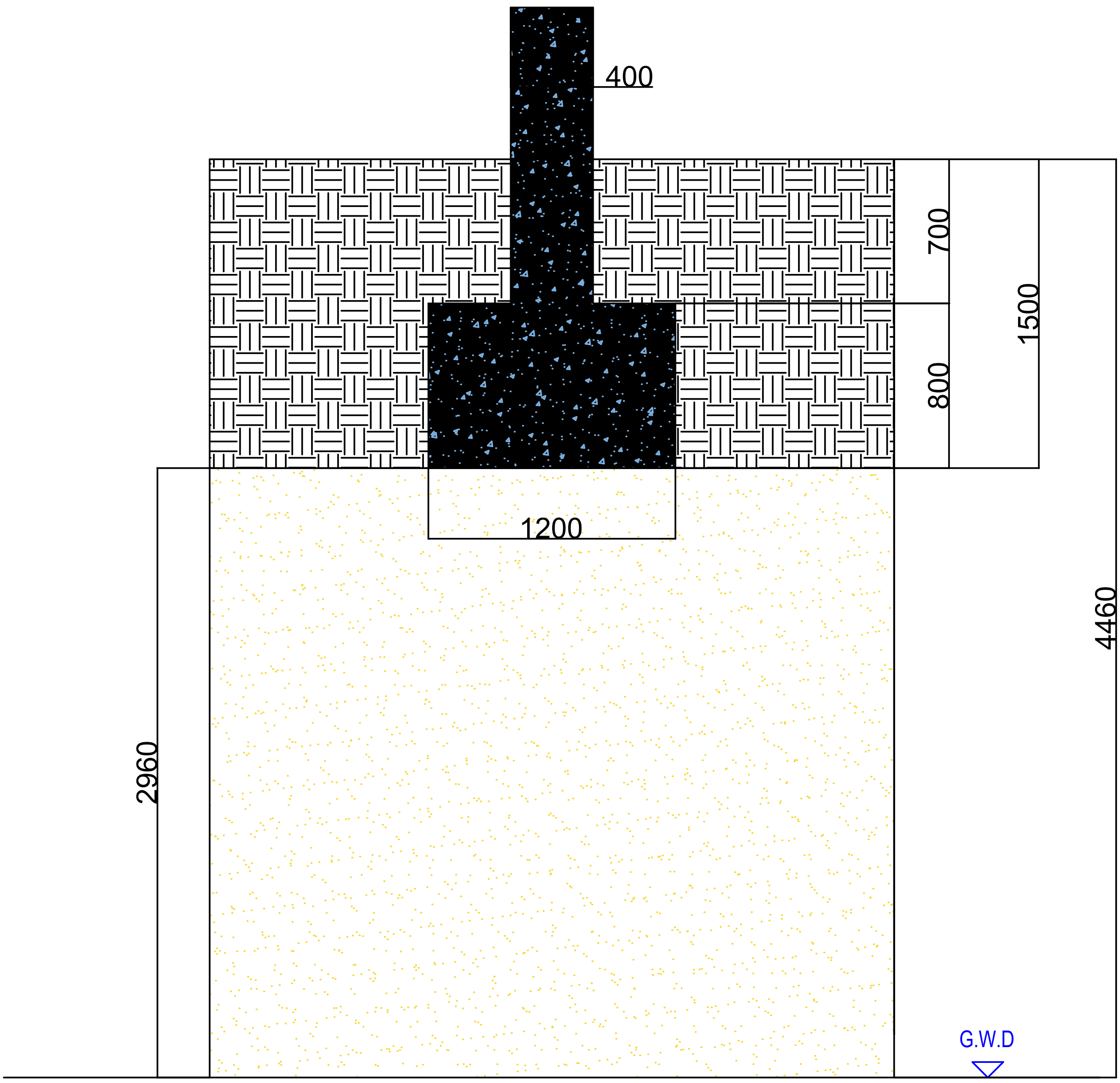
$$\boxed{S_c := \frac{\alpha}{9 \cdot E_c} \cdot q_a \cdot \lambda_c \cdot B = 22.6508\ \text{mm}}$$

$$\boxed{S_d := \frac{2}{9 \cdot E_d} \cdot q_a \cdot B_0 \cdot \left(\lambda_d \cdot \frac{B}{B_0} \right)^{\alpha} = 22.4576\ \text{mm}}$$

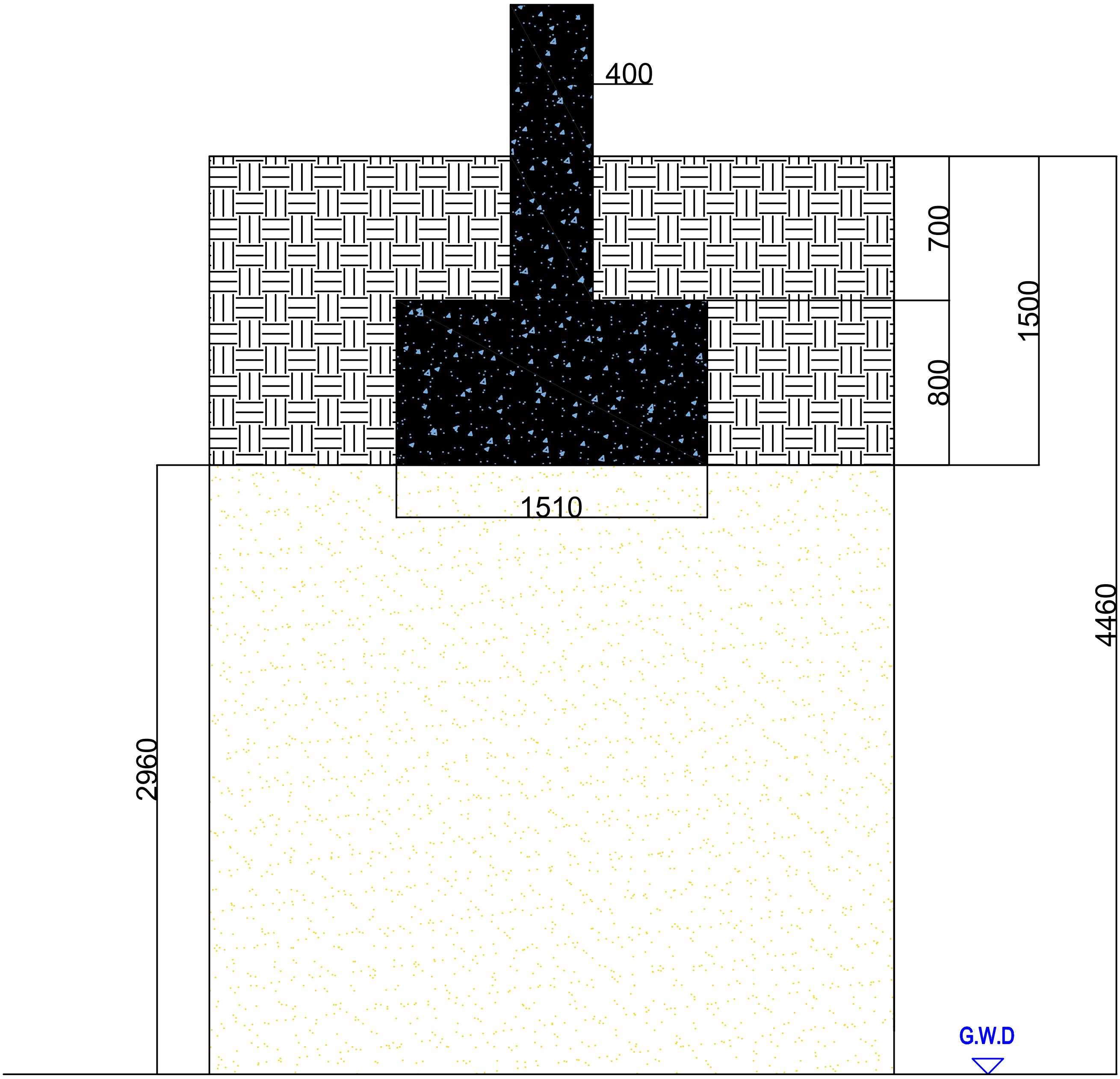
$$\boxed{S_f := S_c + S_d = 45.1083\ \text{mm}}$$

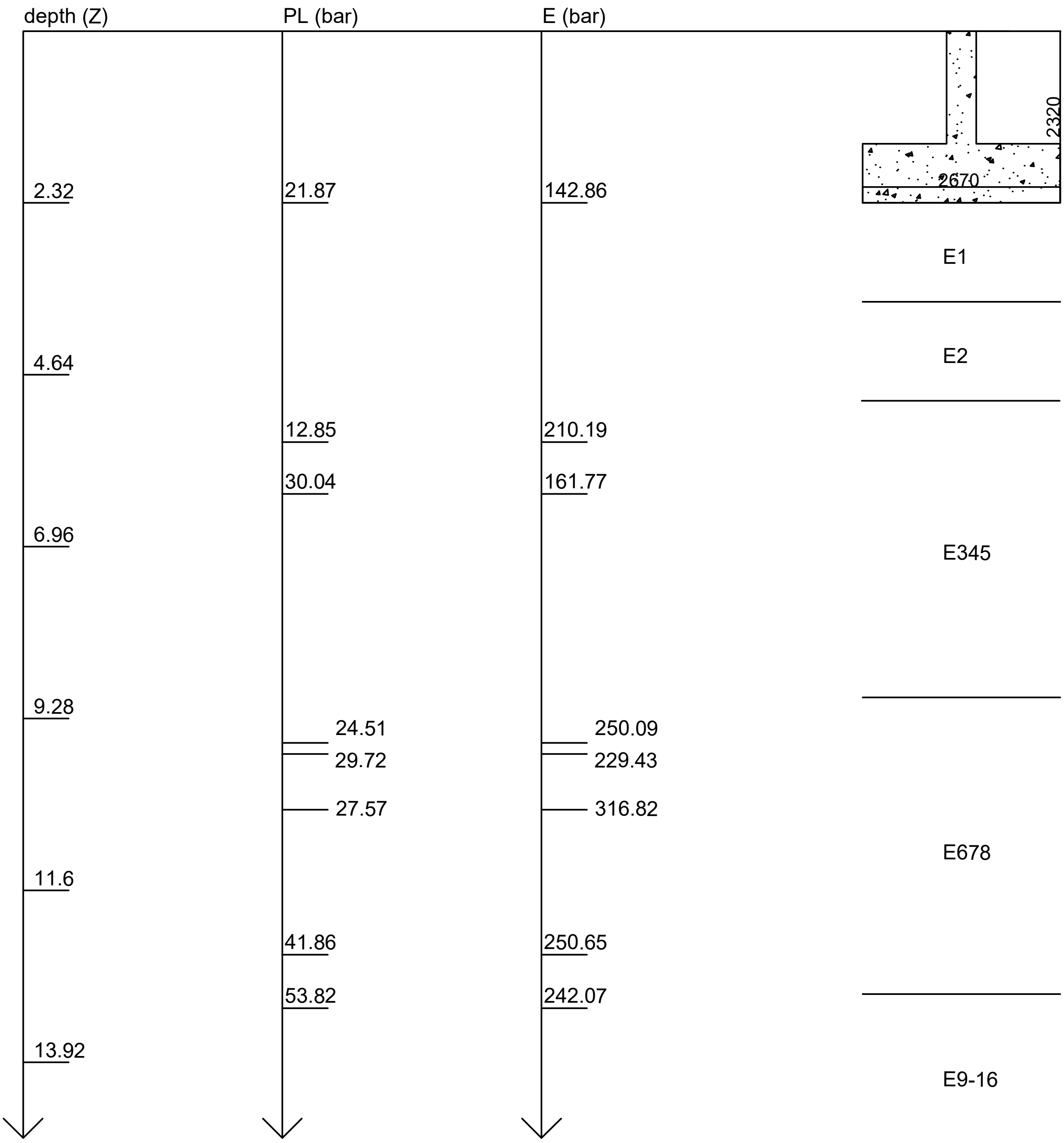
$$S_f > 40\ \text{mm}$$

here we can see that this foundation isn't good since S surpassed the maximam allowable settlement which is 40mm rectangle isolated footing



wall footing




depth (Z)
PL (bar)
E (bar)
2320
2670
2.32
21.87
142.86
E1
E2
4.64
12.85
210.19
30.04
161.77
6.96
E345
9.28
24.51
29.72
27.57
250.09
229.43
316.82
E678
11.6
41.86
250.65
53.82
242.07
13.92
E9-16